

Laura Cuesta Bret

+1 386 843 6008 | lauracuestabret@gmail.com | <https://linkedin.com/in/laura-cuesta-bret-098899293>

EDUCATION

Embry-Riddle Aeronautical University
Bachelor of Science, Aerospace Engineering
Area of concentration: Aeronautics

Daytona Beach, FL
May 2027
GPA: 3.97/4.0

WORK EXPERIENCE

Research Assistant - NASA ULI Safety-Aware Learning and Assured Autonomy for Aviation Applications *April 2026 – Present*

Embry-Riddle Aeronautical University

Daytona Beach, FL

- Conduct indoor flight testing of UAVs to evaluate flight performance and validate autonomous control algorithms
- Tune PID flight controllers in MATLAB/Simulink to improve stability and performance
- Developed a physics-based ground-contact model in Simulink using spring-damper and friction dynamics to enable realistic landing behavior and prevent terrain penetration.
- Integrate and validate autonomous flight-control architectures using MATLAB/Simulink, PX4, ROS 2, Raspberry Pi, Python, and QGroundControl.

AE 307 Grader: Incompressible Aerodynamics

Jan. 2026 – Present

- Grade homework assignments, assessing analytical solutions in incompressible aerodynamics.
- Provide concise feedback to reinforce core aerodynamic concepts and problem-solving methods.

PROJECT EXPERIENCE

ERAU Women's BAJA SAE Team

- Design suspension and brake components using SolidWorks and validate designs with ANSYS static structural analysis.
- Manufacture components and gain hands-on experience with MIG and TIG welding.

Launch Vehicle Project

- Designed a two-stage launch vehicle capable of delivering a 4,700 lb payload to Low Earth Orbit using MATLAB.
- Performed trajectory analysis, mass estimation, and thrust-to-weight calculations in MATLAB and Excel.
- Selected propulsion systems, designed liquid propellant tanks, and developed CAD models in CATIA.
- Collaborated with a five-member engineering team throughout the design process.

Glider Project

- Designed and optimized a balsa wood glider using aircraft design principles, Excel, and Aery32 aerodynamic analysis.
- Manufactured and flight-tested the design to evaluate performance against competing student aircraft.

TECHNICAL SKILLS

- **Engineering software:** CATIA v.5, SIEMENS NX, SolidWorks, Python, MATLAB, SIMULINK, ANSYS, Feemap, PX4, ROS2, QGroundControl
- **Office Software:** PowerPoint, Microsoft Word, Excel
- **Languages:** Spanish (Native), English (fluent)

LEADERSHIP & INVOLVEMENT

- Team Captain – NCAA Division II Women's Tennis *2025 - present*
- Student-Athlete Advisory Committee representative (SAAC) *2025 - present*
- ERAU Honors Program *2023 - present*
- ERAU Women's Baja Club - Suspension *2024 - 2025*